How do 15-to-18-month-olds represent unseen objects when such objects are being named?

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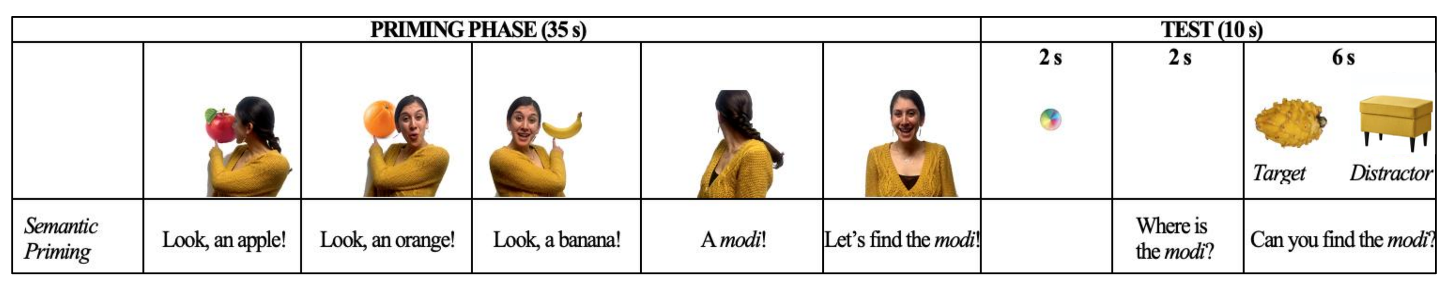
BACKGROUND

- Language helps us learn about things that we have not experienced directly
- This capacity is a hallmark of our species
- It shapes our cultural learning and education practices, clinical interventions
- When does communication about the unseen develop?
- Prior work: 15-month-olds can form a representation of an unseen object, link it to a novel word, and identify it when it becomes visible (Luchkina & Waxman, 2025)



- This ability forms a foundation for further advances in communicating about the unseen, helping us form, retrieve, and modify representations of things we have not experienced perceptually
- **But how do infants build representations of the unseen?**

METHOD

- **Participants:** *N*=162 15-to-18-month-olds (M=16.72) on Children Helping Science
- **Procedure:** An experimenter first labeled three objects familiar to the child, all from the same category (e.g., fruits), and then labeled an object that was apparently visible to her but not to the child (e.g., 'Wug'; Figures 1a and 1b)
- **Manipulated variables:**
  - Priming mode: Visual and Lexical vs. Lexical Only (within subject)
  - Breadth (visual) of the Primed Category: Broad vs. Narrow category (between subjects)
  - Typicality of the Target Object at Test: Typical vs. Atypical exemplar (between subjects)
- **Test:** The child was shown a pair of novel objects side by side: one belonged to the primed category, and the other was unrelated. The child heard a prompt to look at the previously nonvisible object labeled by the experimenter (Figure 1c)
- **Measure:** % of looking time to the target object 367-2000 ms after the onset of the target word (Figure 2)

METHOD (continued)

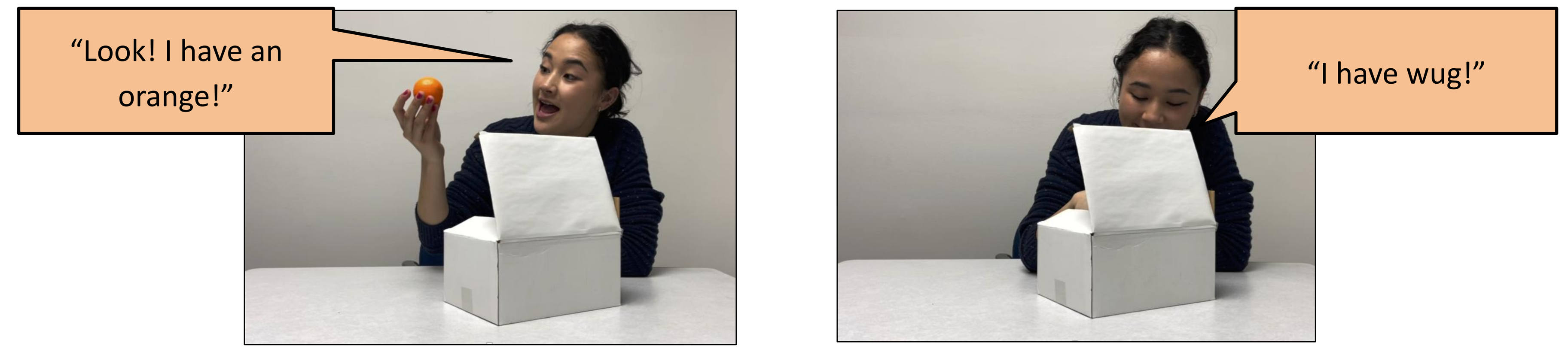


Figure 1a. Familiarization in the Visual and Lexical priming mode. Familiar objects are visible to the child, while the novel one remains hidden. In the Narrow Primed Category condition, all visible objects were similarly shaped (e.g., orange, apple, plum). In the Broad Primed Category condition, the shape of visible objects varied more (e.g., pear, grapes, banana).

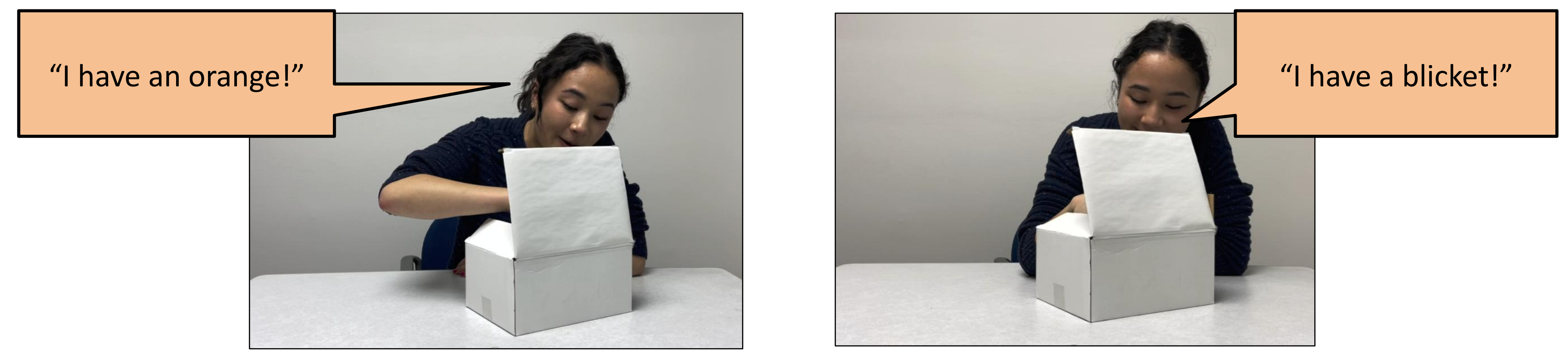


Figure 1b. Familiarization in the Lexical Only priming mode. Familiar objects are visible to the child, while the novel one remains hidden (the novel label is different here because priming mode was manipulated within-subject).

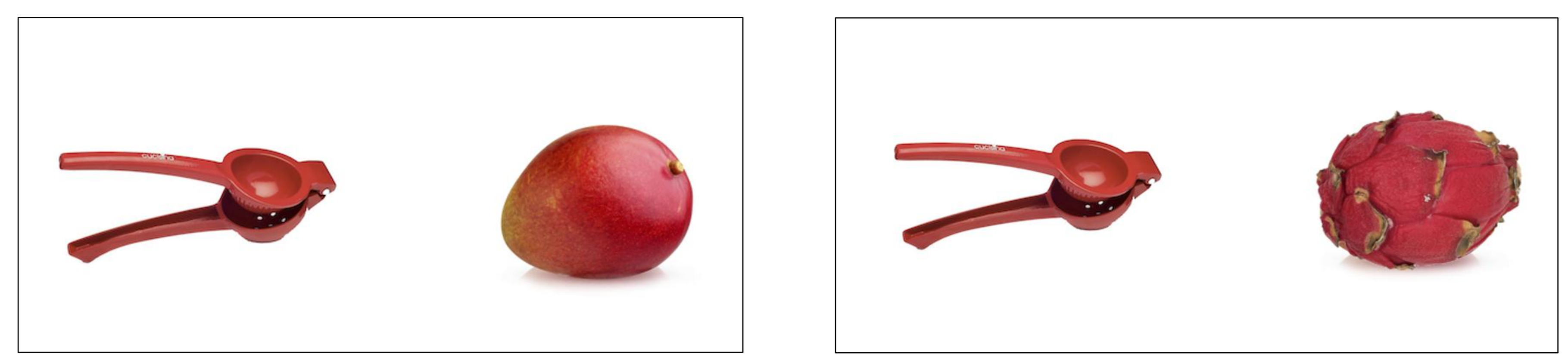


Figure 1c. A test trial, accompanied by a verbal prompt, "Look! A wug/blicket! Do you see the wug/blicket?"
The target was either Typical (i.e., visually similar to the priming objects; left) or Atypical (visually dissimilar to the priming objects; right)

RESULTS

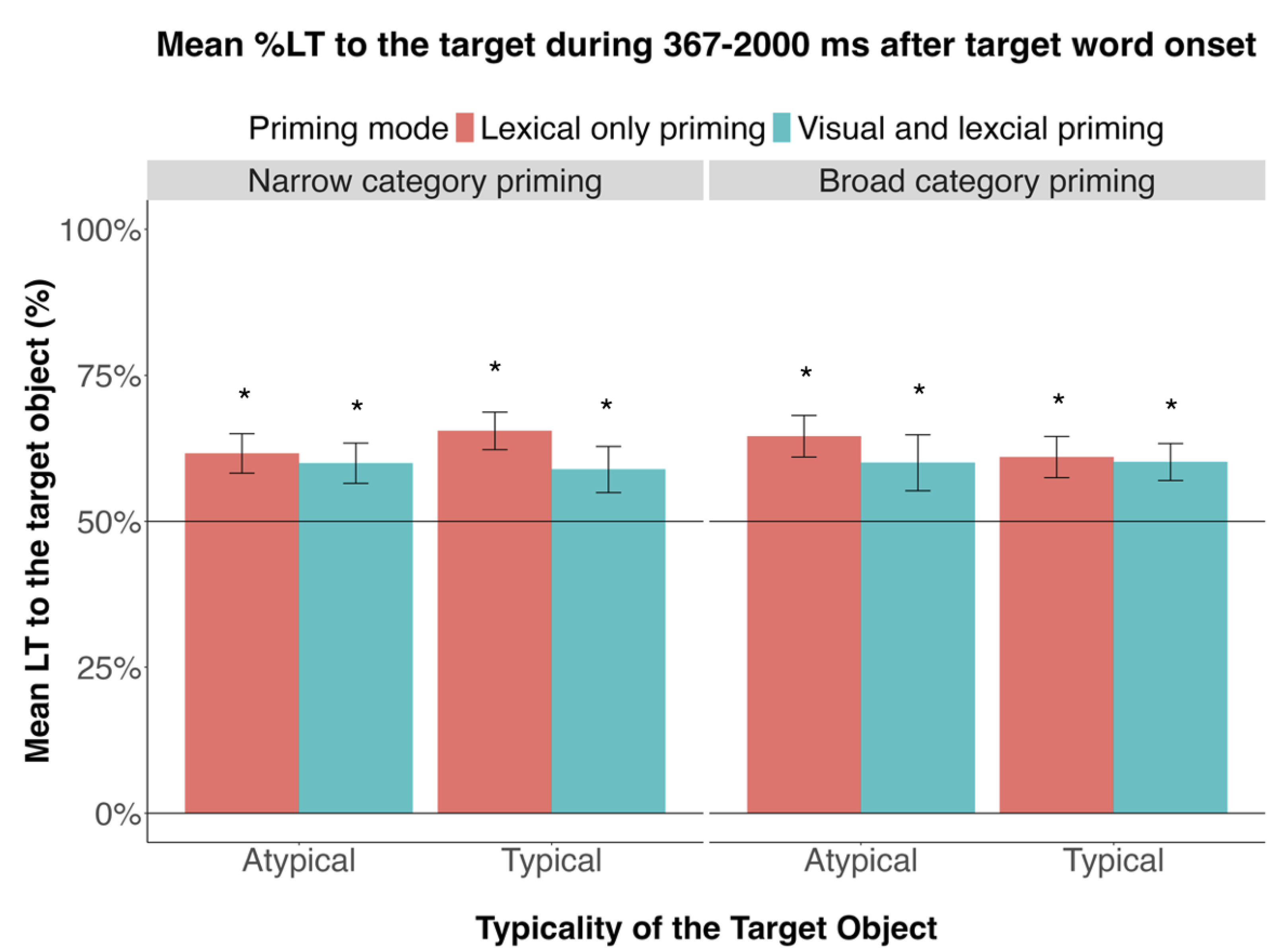


Figure 2. Mean % of looking time to the target object, averaged across the window of analysis

PREDICTIONS AND ANALYSES

- **Predictions:** Visual and Lexical priming and Typical exemplars (especially under Narrow priming) would yield stronger preferences for the target object
- **Analyses: Multivariate ANOVA**
- **Results:**
  - Significant preference for the target in all cases
  - No main effect of Category Breadth
  - No main effect of Typicality of the target
  - No simple effects of priming mode
  - BUT: Main effect of Priming Mode (infants performed better in the Lexical Only mode)

CONCLUSIONS

- Labels alone may be more effective, possibly because:
  - identifying familiar exemplars during priming taxes attention
  - visual input narrows representations, limiting later recognition
  - Infants can form representations of unseen objects from linguistic input alone
- Labels contribute to the development of representational and referential capacities

LIMITATIONS AND NEXT STEPS

- Limitations: Because visible objects are introduced at test, we cannot directly evaluate which specific features infants represented for the unseen object before seeing it
- Next steps: Explore methods that would allow us to evaluate the features (e.g., introduce multiple target object options at test, alongside distractor objects that may share features with the target but belong to different categories), measure looking trajectories

REFERENCES

Luchkina, E., & Waxman, S. (2025). Semantic priming supports infants' ability to learn names of unseen objects. *PLOS ONE, 20*(4), e0321775.